

# Meeting Summaries of July 7

**Date:** July 7, 2026

**Time:** 1:30 - 3:30 Vancouver

**Place:** TRU, S203

**Attendees:** Feng, Prof. Javed, Prof. Yue

**Topic:** In-person meeting. Feng reports the next propotion of his work after last meeting; questioning and feedback from the advisors; discussion on improvements and next steps.

## Checklist of Accomplishments from Last Meeting

- The unit misuse with Al, Fe and Ca elements have been fixed, their concentration records are now in mg/g unit and it updated the downstream results.
- The phrases of talking retained loadings in **L** has been improved, accordingly the summation formula for  $SumRel_i$  is simplified with continuously increasing subscripts without segmentation.
- Two references are cited to support why excluding PC with high loadings on natural elements.

## Questions, Feedbacks and Suggestions

- It is not sufficient to finalize a p% cut-off value for the lowest scores sites as reference (ecological) condition by a hypothesis test that compares whether the lowest p% scored sites are significantly less impacted than a random p% scored sites from all sample units.

Specifically, it uses the stressor **S** and taxa (invertebrate) **T** matrices of a subset  $m_b$  to fit a multivariate regression for constrained taxa matrix  $\hat{\mathbf{T}} = \mathbf{S}\beta$  and output the adjusted  $R^2$  value by  $\bar{R}^2_{m_b} = adj\left(\frac{tr(\hat{\mathbf{T}}^T\hat{\mathbf{T}})}{tr(\mathbf{T}^T\mathbf{T})}\right)$  to reflect how much of the community variation can be explained by the stressors **S**.

$$f : (\mathbf{S}_{m_b}, \mathbf{T}_{m_b}) \rightarrow \hat{\mathbf{T}}_{m_b} \rightarrow \frac{tr(\hat{\mathbf{T}}^T\hat{\mathbf{T}})}{tr(\mathbf{T}^T\mathbf{T})} \rightarrow \bar{R}^2_{m_b}$$

Create a set  $m_{ref}$  with the lowest p% scores sites and a set  $m_b$  with the same number of random sites from all sample units, there are  $B$  bootstrapped sets  $\{m_b\}_1^B$ . Initialize a vector as  $[m_1, m_2, \dots, m_B]^T$ , apply the  $f$  function entry-wise to  $M$  to map these entry sets to  $\bar{R}^2_{m_b}$  values:

$$f_{broadcast} : [m_1, m_2, \dots, m_B]^T \rightarrow [\bar{R}^2_{m_1}, \bar{R}^2_{m_2}, \dots, \bar{R}^2_{m_B}]^T$$

The resulted vector of  $\bar{R}^2_{m_b}$  values portraits the null distribution of  $\bar{R}^2$  for a random p%-size set with  $B$  **bootstrapped samples**. However, the issue is that the  $\bar{R}^2_{m_b;p\%}$

of a random  $p\%$  size set  $m_b$  converges to a normal distribution of the  $\bar{R}_m^2$  of the overall sampled units.

$$\sqrt{p\%m}(\bar{R}_{m_b}^2 - \bar{R}_m^2) \xrightarrow{d} N(0, \sigma^2)$$

It favors a scenario that the larger the  $p\%$  size, the more concentrated the random  $\bar{R}_{m_b}^2$  values are around the overall sampled  $\bar{R}_m^2$ . Consequently, this hypothesis test can only support where to choose a smallest  $p\%$  cut-off value for a significantly less impacted set, but it **cannot** tell where to stop the  $p\%$  cut-off value.

## Action Items

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- Feng needs to fix these mentioned issues, bring the fixes and prepare the next proposition for the next meeting.

## Summary of Discussion

- Just normal report as usual.

## Next Meeting Time

- **Date and Time:** July 9, 2026, 8:00 - 9:00 Vancouver
- **Form:** Online, zoom meeting.